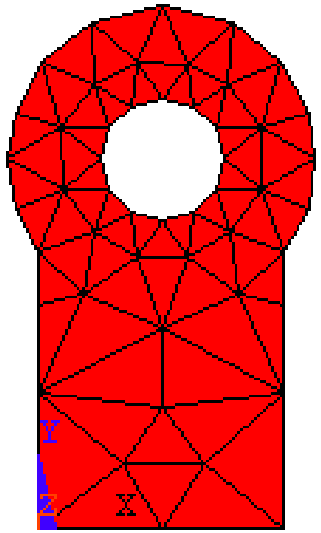
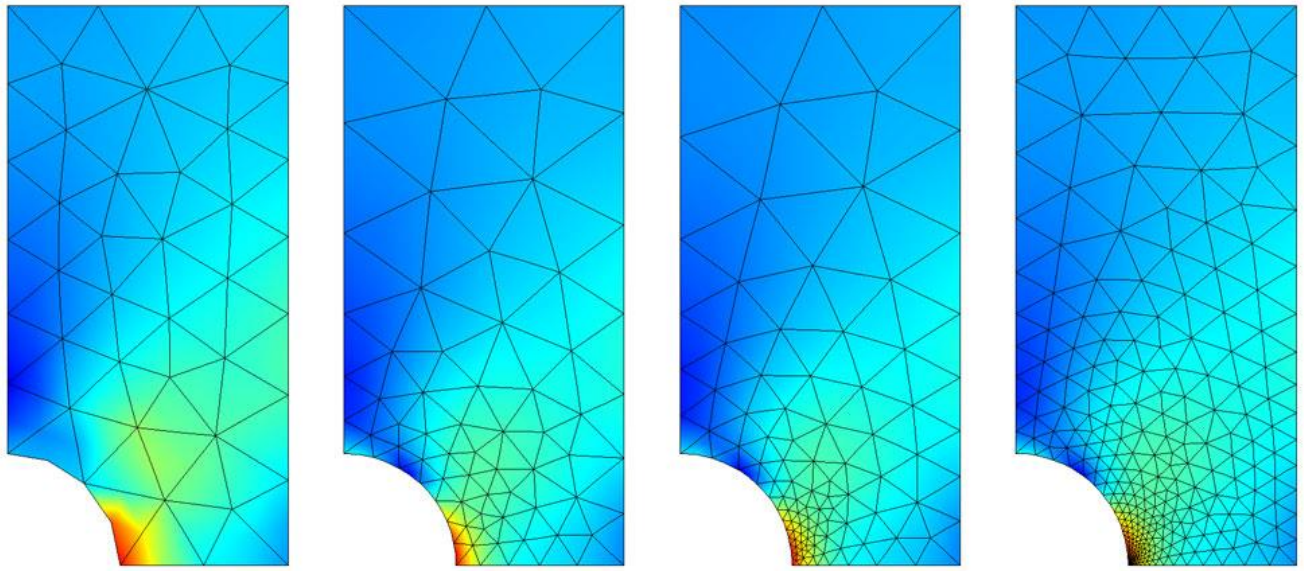
ESA 321

Lecture 6 - Plane Element &
Constant Strain Triangle
(CST)



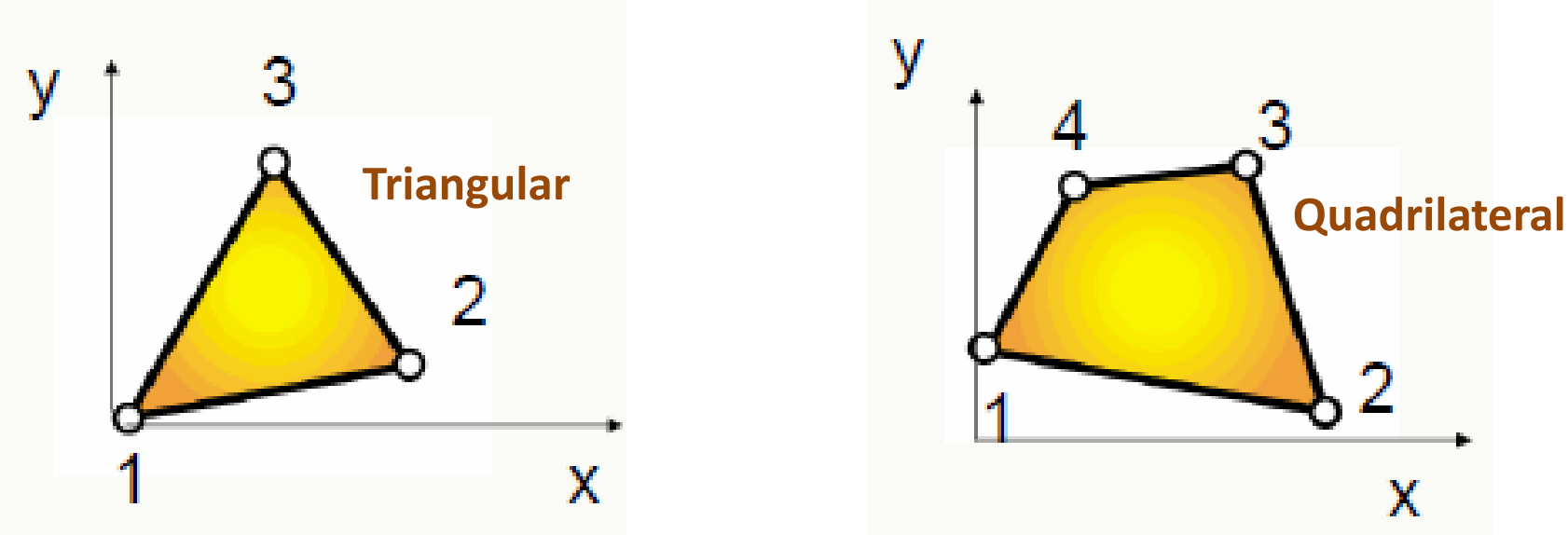
Objectives

- Able to apply FEM using Constant Strain triangle element for solving structural problems.
- Able to use Finite Element Software to perform FE analysis **(LISA 8.0.0)**



Area Element / Planar Element

- 2D planar elements (**CST, LST, QUAD, 8 node Quad**) are **thin-plate elements**.
- They're (usually) utilised, when a 3D problem can be simplified into a 2D problem.

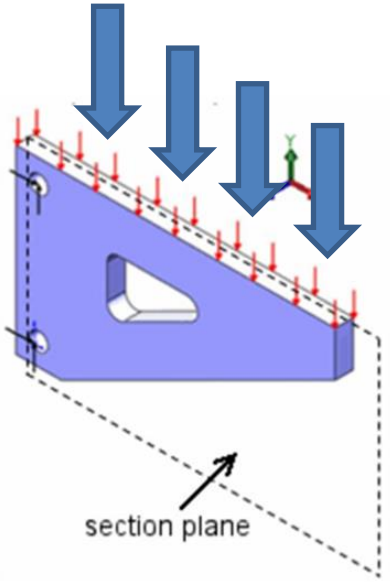


Example of 2D elements

Area Element / Planar Element

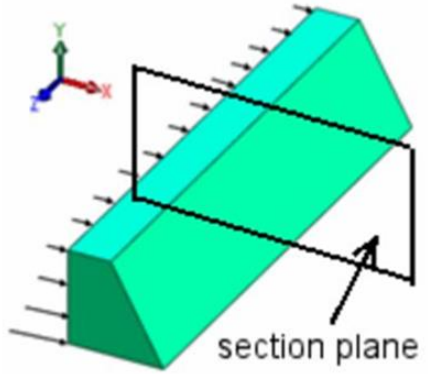
The 2D planar elements are extremely important for:

1. **Plane stress analysis:**
   - **Plane stress is a condition in which stresses in a structure are confined to a single plane such as when a thin structure is subjected to loads acting in one plane** resulting in local stress concentrations.



Plane Stress

2. **Plane strain analysis:**
   - **Plane strain is a condition in which strains in a structure are confined to a single plane. Such as when a long structure is subjected to uniform loading over the structure's entire length.**

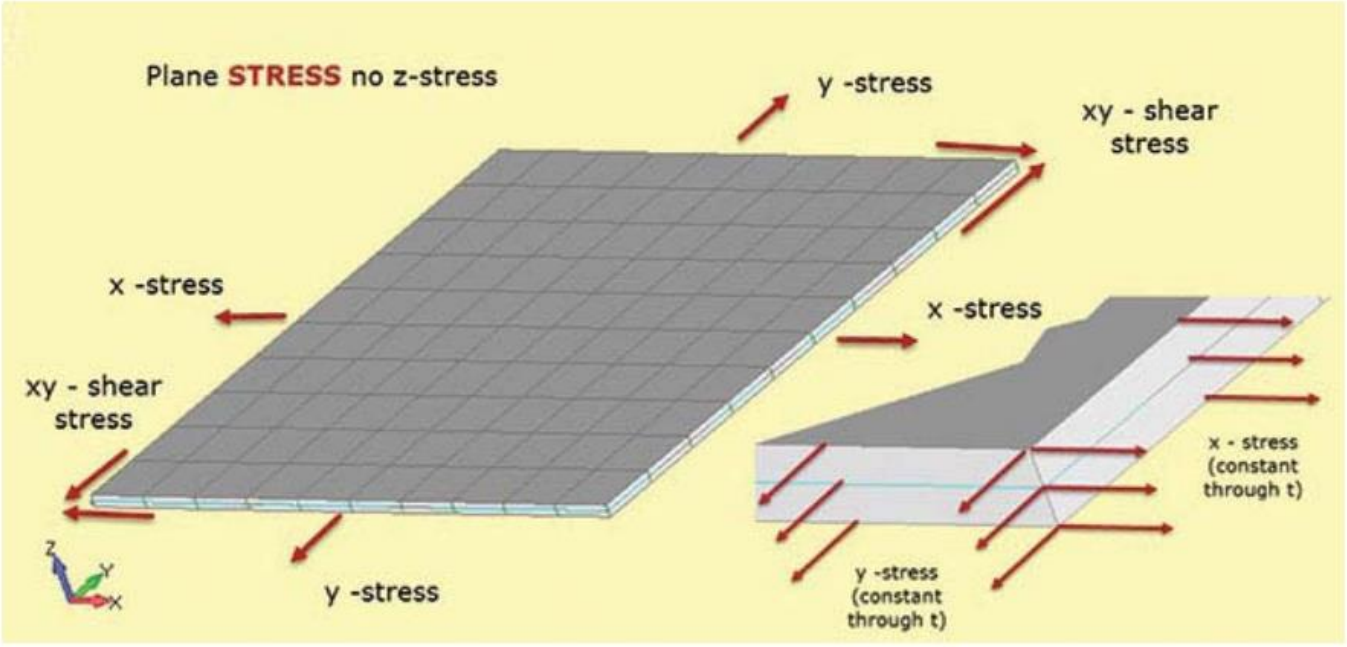


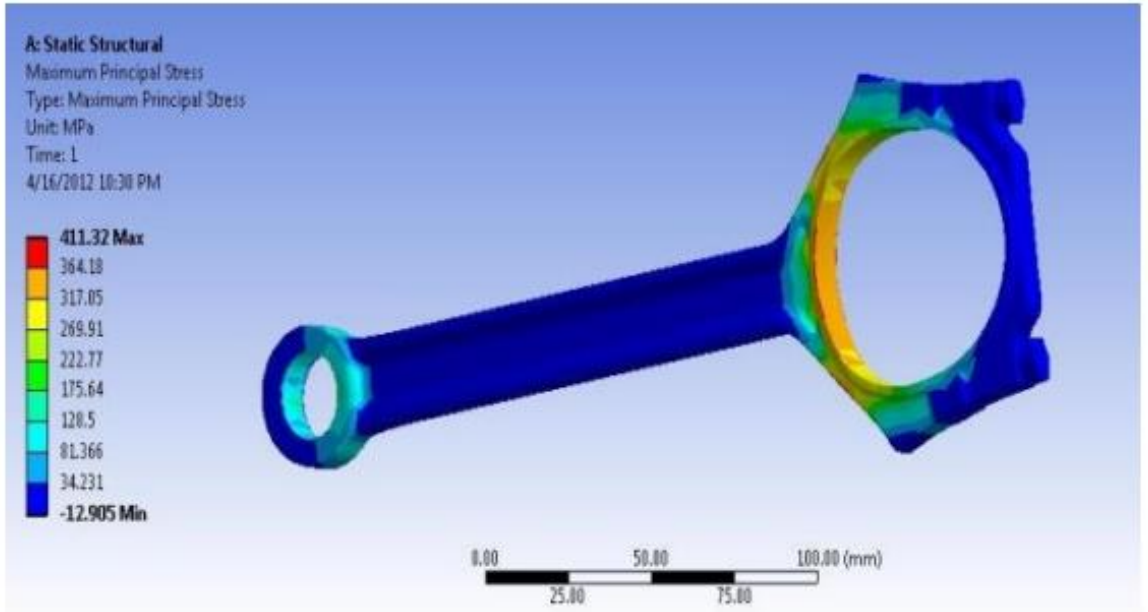
Plane Strain

Classification of Plane Stress or Plane Strain analysis always requires engineering judgement.

Area Element / Planar Element

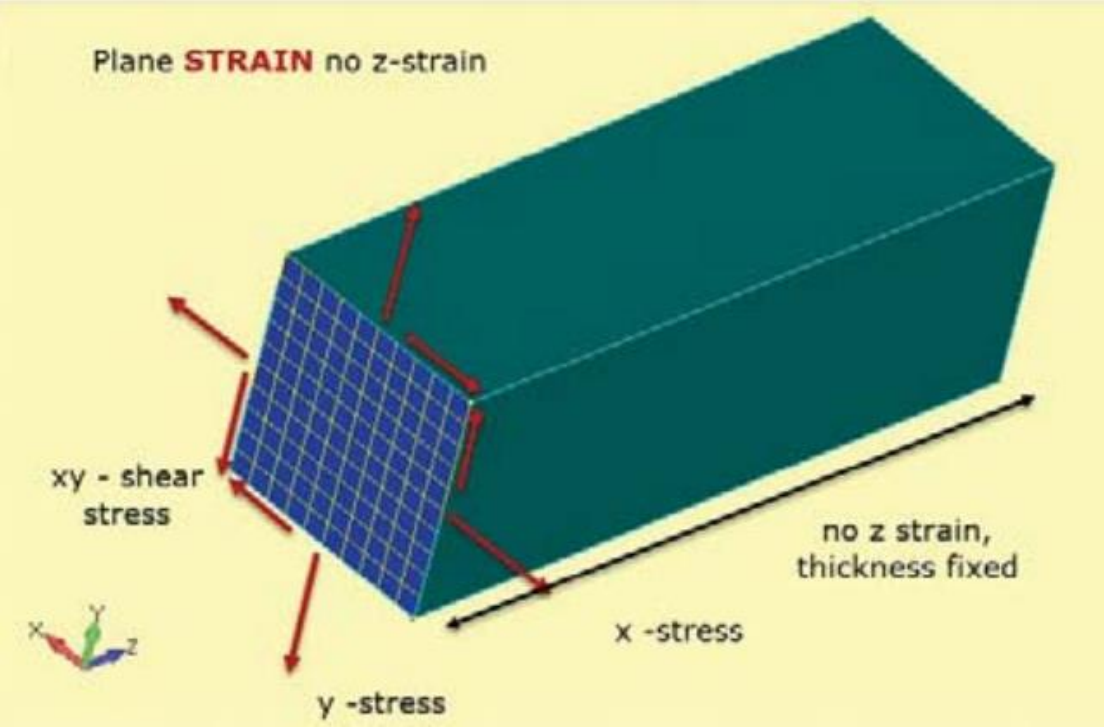
- **Plane stress** occurs when **all of the stresses** acting on the components are **in the same plane**. Stresses perpendicular to the plane is minimal and can be neglected.
- Applicable for **thin structures** where **stresses variation** between top and bottom surface is **minimum**. E.g. Piston conrod.

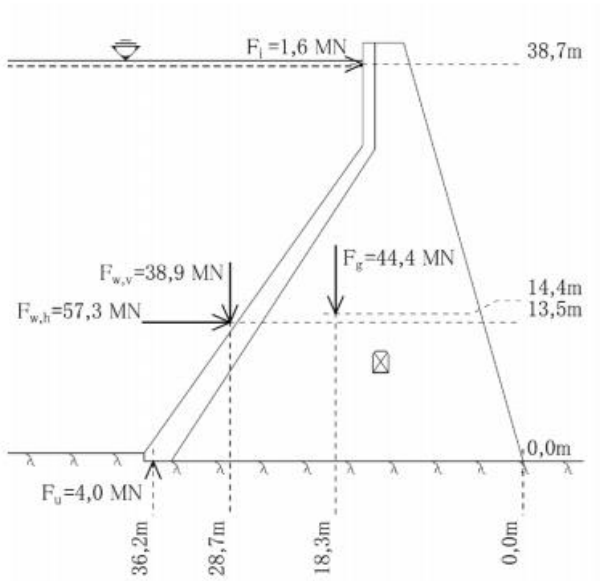




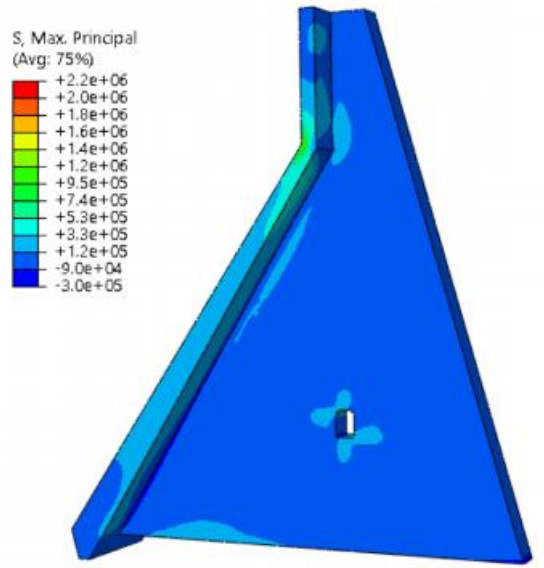
Area Element / Planar Element

- **Plane strain** occurs when the strain normal to the x-y plane $\in_z$ and the shear strains γ_{xz} and γ_{yz} are assumed to be zero. **Use the cross section for analysis**. E.g. dam wall, box culvert.





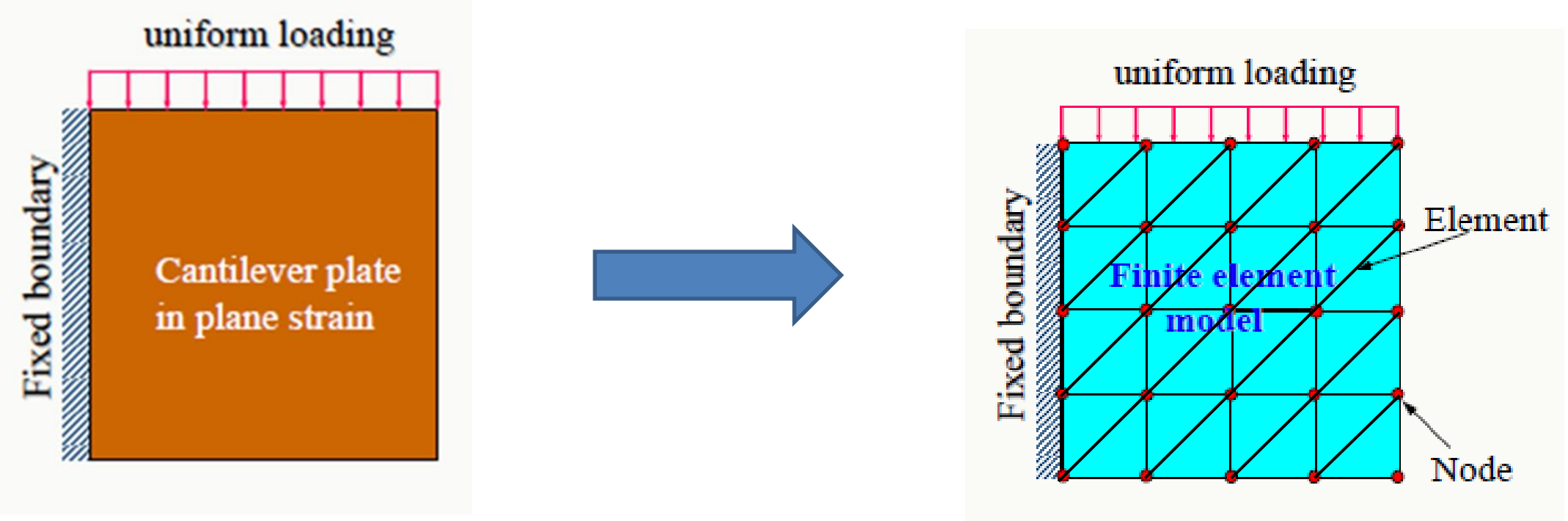
a)



b)

Constant Strain Triangle

– The two-dimensional body is discretized into finite element model by subdividing it using constant strain triangle elements.

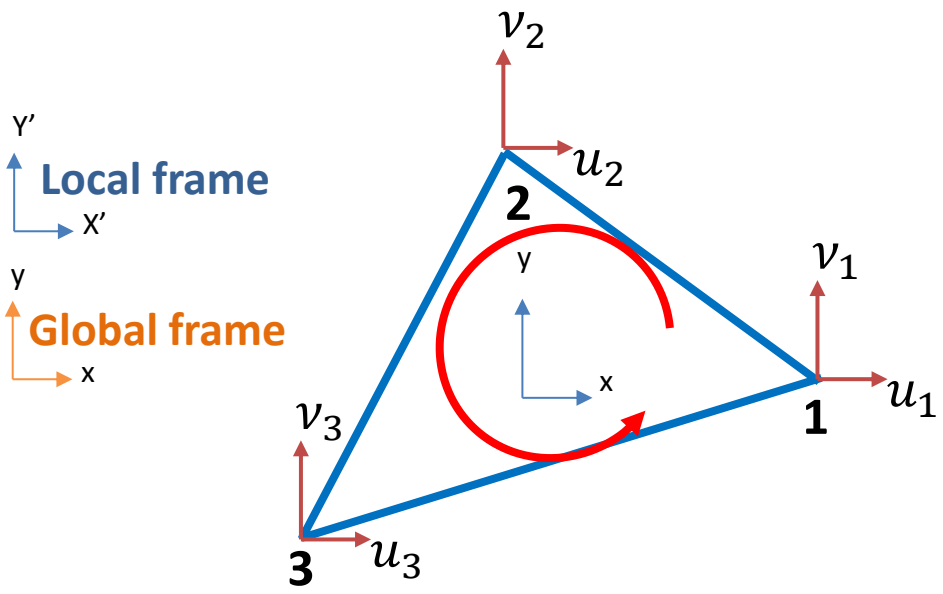


CST Formulation

- CST is a 3 nodes triangular element.
- Each node has 2 dofs.
- There are a total of 6 dofs for a each element, [k]e will be a (6 x 6) rectangular stiffness matrix for each element.

$$\begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_{x1} \\ F_{y1} \\ F_{x2} \\ F_{y2} \\ F_{x3} \\ F_{y3} \end{Bmatrix}$$

$[K]^e$ **Stiffness Matrix for one element**



CST Stiffness Matrix

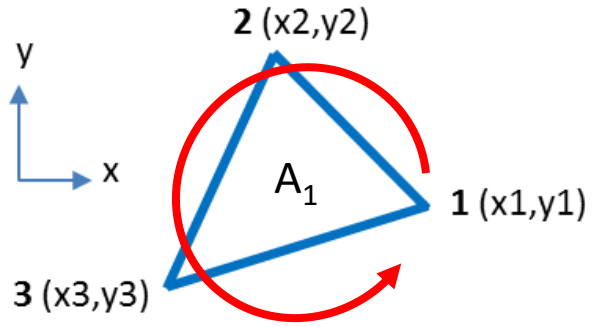
Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

$$A_1 = \frac{1}{2}\left|\det[J]\right|$$

[J] - Jacobian matrix

$$[J] = \begin{bmatrix} x_{13} & y_{13} \\ x_{23} & y_{23} \end{bmatrix}$$



$$\det[J] = x_{13}y_{23} - x_{23}y_{13}$$

Coordinate $X_{13} = X_1 - X_3$

CST Stiffness Matrix

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

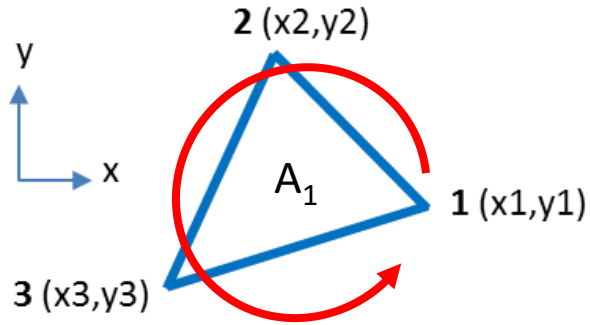
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$$\det[J] = x_{13}y_{23} - x_{23}y_{13}$$

Coordinate $X_{13} = X_1 - X_3$



[B] matrix is known as the strain-displacement matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$

Coordinate $X_{32} = X_3 - X_2$

CST Stiffness Matrix

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

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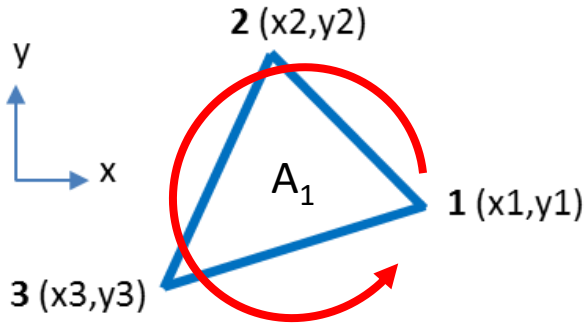
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Coordinate $X_{32} = X_3 - X_2$



[D] matrix is known as the material matrix

Plane Stress

$$[D] = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1-\nu) \end{bmatrix}$$

Plane Strain

$$[D] = \frac{E}{(1+\nu)(1-2\nu)} \cdot \begin{Bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1}{2}-\nu \end{Bmatrix}$$

CST Stiffness Matrix

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B] = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix}$$

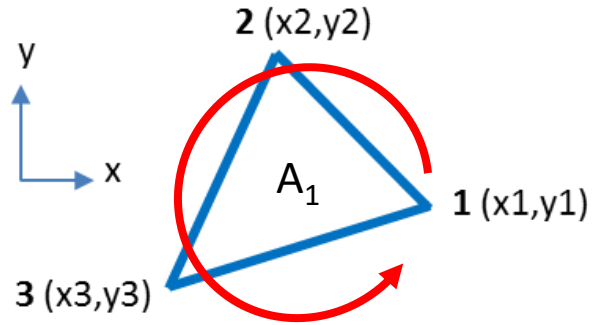
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Coordinate $X_{32} = X_3 - X_2$

$$A_1 = \frac{1}{2}\left|\det[J]\right|$$

[J] - Jacobian matrix



$$[J] = \begin{bmatrix} x_{13} & y_{13} \\ x_{23} & y_{23} \end{bmatrix}$$

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$$\det[J] = x_{13}y_{23} - x_{23}y_{13}$$

Plane Stress

$$[D] = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1-\nu) \end{bmatrix}$$

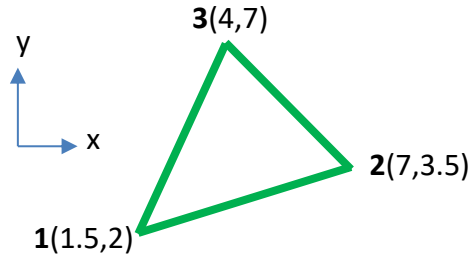
Plane Strain

$$[D] = \frac{E}{(1+\nu)(1-2\nu)} \cdot \begin{Bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1}{2}-\nu \end{Bmatrix}$$

Coordinate $X_{13} = X_1 - X_3$

Example

Determine the **element stiffness matrix** for the straight sided triangular element shown. The **thickness is 1 mm** and ***E* is 70 GPA** with **v = 0.3**. Assume **plane stress** condition.

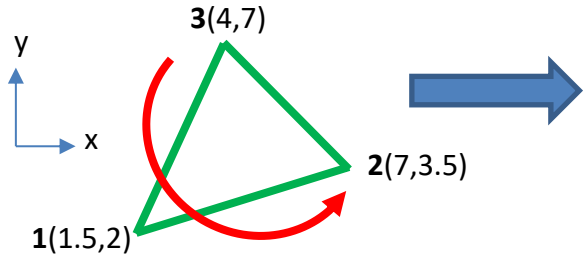


Coordinate in mm

Plane stress

Example

Determine the **element stiffness matrix** for the straight sided triangular element shown. The **thickness is 1 mm** and ***E* is 70 GPA** with **v = 0.3**. Assume **plane stress** condition.



Coordinate in mm

Plane stress

x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

t = 1mm

E = 70 GPa

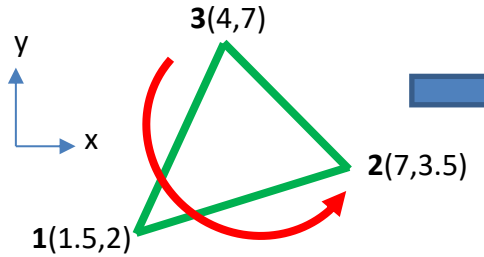
V = 0.3

Example

Determine the **element stiffness matrix** for the straight sided triangular element shown. The **thickness is 1 mm** and ***E* is 70 GPA** with **v = 0.3**. Assume **plane stress** condition.

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$



Coordinate in mm

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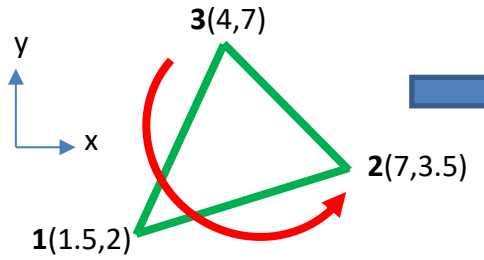
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$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

$t_1 = 1mm$



Coordinate in mm

Plane stress

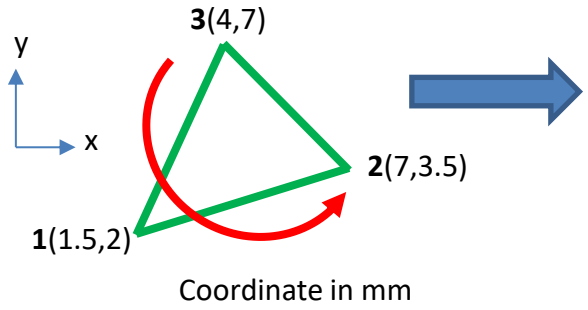
x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

t = 1mm
E = 70 GPa
V = 0.3

We find area of the triangle

Example

Determine the **element stiffness matrix** for the straight sided triangular element shown. The **thickness is 1 mm** and ***E* is 70 GPA** with **v = 0.3**. Assume **plane stress** condition.



Plane stress

x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

$t_1 = 1mm$

t = 1mm
E = 70 GPa
V = 0.3

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

We find area of the triangle

$$A_1 = \frac{1}{2}|\det[J]| \qquad [J] = \begin{bmatrix} x_{13} & y_{13} \\ x_{23} & y_{23} \end{bmatrix}$$

$$\det[J] = x_{13}y_{23} - x_{23}y_{13}$$

$$x_{13} = x_1 - x_3 = 1.5 - 4 = -2.5$$

$$y_{23} = y_2 - y_3 = 3.5 - 7 = -3.5$$

$$x_{23} = x_2 - x_3 = 7 - 4 = 3$$

$$y_{13} = y_1 - y_3 = 2 - 7 = -5$$

$$det[J] = |(-2.5)(-3.5) - (3)(-5)|$$

$$= |(8.75) - (-15)|$$

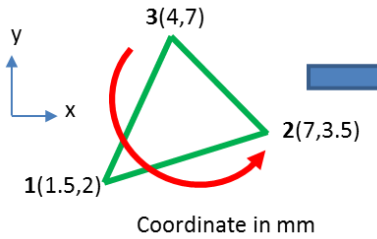
$$= \mathbf{|23.75|}$$

$$A_1 = \frac{1}{2}|23.75|$$

$$= \mathbf{11.875\ mm^2}$$

Area must always be positive

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

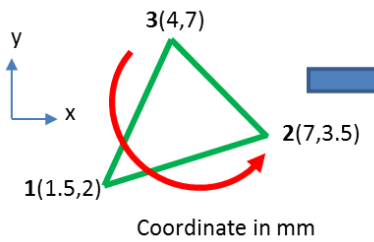
$$det[J] = |23.75|$$

$$A = 11.875\ mm^2$$

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Strain-Displacement Matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$



Coordinate in mm

x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

$$det[J] = |23.75|$$

$$A = 11.875\ mm^2$$

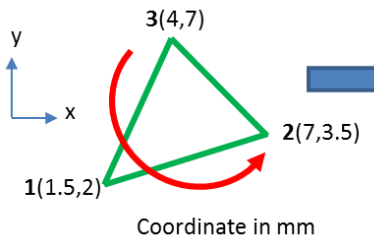
$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Strain-Displacement Matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$

$$= \frac{1}{23.75} \begin{bmatrix} 3.5-7 & 0 & 7-2 & 0 & 2-3.5 & 0 \\ 0 & 4-7 & 0 & 1.5-4 & 0 & 7-1.5 \\ 4-7 & 3.5-7 & 1.5-4 & 7-2 & 7-1.5 & 2-3.5 \end{bmatrix}$$

$$[B] = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$



Coordinate in mm

x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

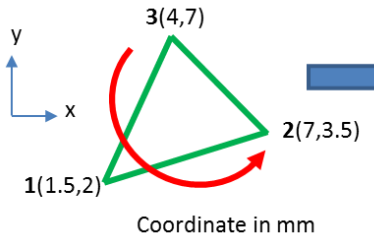
E = 70 GPa
V = 0.3
t = 1mm

$$det[J] = |23.75|$$
$$A = 11.875\ mm^2$$

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Strain-Displacement Matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

$$det[J] = |23.75|$$
$$A = 11.875\ mm^2$$

$$= \frac{1}{23.75} \begin{bmatrix} 3.5-7 & 0 & 7-2 & 0 & 2-3.5 & 0 \\ 0 & 4-7 & 0 & 1.5-4 & 0 & 7-1.5 \\ 4-7 & 3.5-7 & 1.5-4 & 7-2 & 7-1.5 & 2-3.5 \end{bmatrix}$$

$$[B] = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

$$[B]^T = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix}$$

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

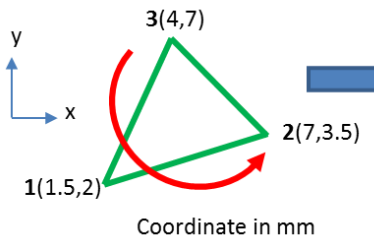
Strain-Displacement Matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$

$$= \frac{1}{23.75} \begin{bmatrix} 3.5-7 & 0 & 7-2 & 0 & 2-3.5 & 0 \\ 0 & 4-7 & 0 & 1.5-4 & 0 & 7-1.5 \\ 4-7 & 3.5-7 & 1.5-4 & 7-2 & 7-1.5 & 2-3.5 \end{bmatrix}$$

$$[B] = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

$$[B]^T = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix}$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

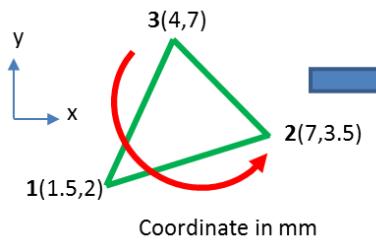
$$det[J] = |23.75|$$
$$A = 11.875\ mm^2$$

Material Matrix

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Strain-Displacement Matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

$$= \frac{1}{23.75} \begin{bmatrix} 3.5-7 & 0 & 7-2 & 0 & 2-3.5 & 0 \\ 0 & 4-7 & 0 & 1.5-4 & 0 & 7-1.5 \\ 4-7 & 3.5-7 & 1.5-4 & 7-2 & 7-1.5 & 2-3.5 \end{bmatrix}$$

$$[B] = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

$$[B]^T = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix}$$

$$det[J] = |23.75|$$

$$A = 11.875\ mm^2$$

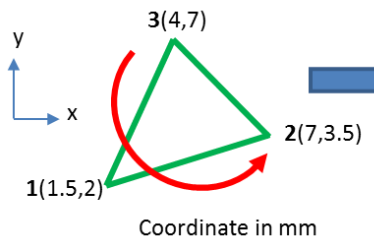
Material Matrix

$$[D] = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1-\nu) \end{bmatrix}$$

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Strain-Displacement Matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

$$= \frac{1}{23.75} \begin{bmatrix} 3.5-7 & 0 & 7-2 & 0 & 2-3.5 & 0 \\ 0 & 4-7 & 0 & 1.5-4 & 0 & 7-1.5 \\ 4-7 & 3.5-7 & 1.5-4 & 7-2 & 7-1.5 & 2-3.5 \end{bmatrix}$$

$$[B] = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

$$[B]^T = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix}$$

$$det[J] = |23.75|$$

$$A = 11.875\ mm^2$$

Material Matrix

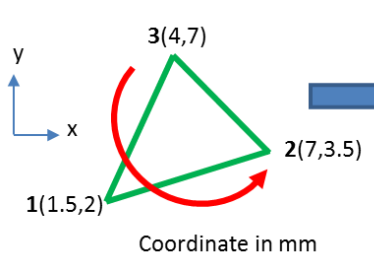
$$[D] = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1-\nu) \end{bmatrix}$$

$$= \frac{70 \times 10^3}{1-0.3^2} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1-0.3) \end{bmatrix}$$

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Strain-Displacement Matrix

$$[B] = \frac{1}{\det[J]} \begin{bmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{bmatrix}$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

$$= \frac{1}{23.75} \begin{bmatrix} 3.5-7 & 0 & 7-2 & 0 & 2-3.5 & 0 \\ 0 & 4-7 & 0 & 1.5-4 & 0 & 7-1.5 \\ 4-7 & 3.5-7 & 1.5-4 & 7-2 & 7-1.5 & 2-3.5 \end{bmatrix}$$

$$[B] = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

$$[B]^T = \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix}$$

$$det[J] = |23.75|$$

$$A = 11.875\ mm^2$$

Material Matrix

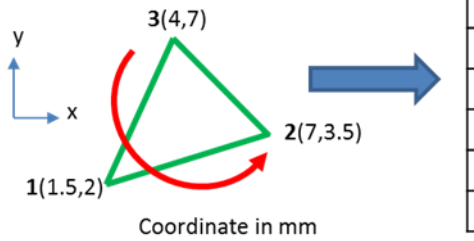
$$[D] = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1-\nu) \end{bmatrix}$$

$GPa(\frac{N}{m^2})$ to $\frac{N}{mm^2}$

$$= \frac{70 \times 10^3}{1-0.3^2} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & \frac{1}{2}(1-0.3) \end{bmatrix}$$

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

$$det[J] = |23.75|$$

$$A = 11.875\ mm^2$$

Multiplication steps - $[B]^T$ ([D] X [B])

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Plane stress $[D]$

t_e A_e $[B]^T$ $[D]$ $[B]$

$$[k]^e = 1 \times 11.875 \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix} \times \frac{70 \times 10^3}{(1-0.3^2)} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

Multiplication steps - $[B]^T$ ([D] X [B])

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Plane stress $[D]$

t_e A_e $[B]^T$ $[D]$ $[B]$

$$[k]^e = 1 \times 11.875 \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix} \times \frac{70 \times 10^3}{(1-0.3^2)} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

u1	v1	u2	v2	u3	v3
2.494	1.105	−2.409	−0.425	−0.085	−0.68
	2.152	−0.233	0.223	−0.873	−2.374
		4.403	−1.316	−1.994	1.549
			2.429	1.741	−2.652
				2.079	−0.868
symmetry					5.026

$$[k]^e = 10^4 \cdot$$

Multiplication steps - $[B]^T$ ([D] X [B])

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$

Plane stress $[D]$

t_e A_e $[B]^T$ $[D]$ $[B]$

$$[k]^e = 1 \times 11.875 \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix} \times \frac{70 \times 10^3}{(1-0.3^2)} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

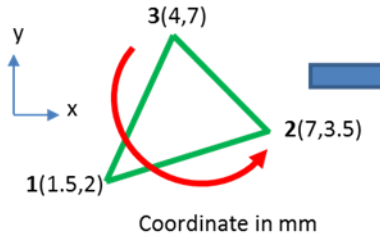
	u1	v1	u2	v2	u3	v3
	2.494	1.105	−2.409	−0.425	−0.085	−0.68
		2.152	−0.233	0.223	−0.873	−2.374
			4.403	−1.316	−1.994	1.549
				2.429	1.741	−2.652
					2.079	−0.868
	symmetry					5.026

$$10^4 \cdot \begin{bmatrix} 2.494 & 1.105 & -2.409 & -0.425 & -0.085 & -0.68 \\ & 2.152 & -0.233 & 0.223 & -0.873 & -2.374 \\ & & 4.403 & -1.316 & -1.994 & 1.549 \\ & & & 2.429 & 1.741 & -2.652 \\ & & & & 2.079 & -0.868 \\ & symmetry & & & & 5.026 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_{x1} \\ F_{y1} \\ F_{x2} \\ F_{y2} \\ F_{x3} \\ F_{y3} \end{Bmatrix}$$

Multiplication steps - $[B]^T$ ([D] X [B])

Element stiffness matrix is given by

$$[k]^e = t_e A_e [B]^T [D] \cdot [B]$$



x1	1.5
y1	2
x2	7
y2	3.5
x3	4
y3	7

Plane stress

E = 70 GPa
V = 0.3
t = 1mm

$t_e \quad A_e \qquad [B]^T \qquad [D] \qquad [B]$

$$[k]^e = 1 \times 11.875 \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & -3 \\ 0 & -3 & -3.5 \\ 5 & 0 & -2.5 \\ 0 & -2.5 & 5 \\ -1.5 & 0 & 5.5 \\ 0 & 5.5 & -1.5 \end{bmatrix} \times \frac{70 \times 10^3}{(1-0.3^2)} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} \times \frac{1}{23.75} \begin{bmatrix} -3.5 & 0 & 5 & 0 & -1.5 & 0 \\ 0 & -3 & 0 & -2.5 & 0 & 5.5 \\ -3 & -3.5 & -2.5 & 5 & -5.5 & -1.5 \end{bmatrix}$$

$$det[J] = |23.75|$$

$$A = 11.875\ mm^2$$

u1	v1	u2	v2	u3	v3	
2.494	1.105	−2.409	−0.425	−0.085	−0.68	u_1
	2.152	−0.233	0.223	−0.873	−2.374	v_1
		4.403	−1.316	−1.994	1.549	u_2
			2.429	1.741	−2.652	v_2
				2.079	−0.868	u_3
symmetry					5.026	v_3

$$10^4 \cdot \begin{bmatrix} 2.494 & 1.105 & -2.409 & -0.425 & -0.085 & -0.68 \\ & 2.152 & -0.233 & 0.223 & -0.873 & -2.374 \\ & & 4.403 & -1.316 & -1.994 & 1.549 \\ & & & 2.429 & 1.741 & -2.652 \\ & & & & 2.079 & -0.868 \\ & symmetry & & & & 5.026 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} F_{x1} \\ F_{y1} \\ F_{x2} \\ F_{y2} \\ F_{x3} \\ F_{y3} \end{Bmatrix}$$

THIS IS FOR ONE CST ELEMENT

Multiplication steps - $[B]^T$ ([D] X [B])

Thank you